Re: Response to New Business Formation and Growth Subindex Evaluation

Dear Professor Mahoney,

Thank you for your thorough and thoughtful evaluation of the New Business Formation and Growth subindex. We sincerely appreciate the care you took in reviewing this element of our Index, and are very pleased you recommend approval as it is. We have made one substantive addition to the subindex based on your recommendation.

I. Overall Index

Education

> *"Will be interested to hear how the authors think about the role of education – perhaps just another societal component that is an important factor driving economic outcomes but outside the capitalist framework?"*

We view education as a critical social endowment that strongly conditions economic performance, but not as a defining institutional component of capitalism itself. While education shapes how effectively capitalist systems operate, it does not constitute a market institution or policy mechanism in the same sense as our eight subindex topics. That said, we do incorporate measures of educational investment, attainment, and quality within the New Business Formation and Growth subindex, where human capital directly affects entrepreneurial capacity and firm creation. In this subindex, education is treated as an enabling input to capitalist activity rather than as a core dimension of capitalism.

Practical PCA Examples

> *"Both the New Business Formation and Growth Overview, and the Technical Appendix are at a rather high level of abstraction/ theoretical level which makes it sound a bit vague, I'd recommend providing one concrete example and walking readers through it. If you want a large number of scholars to engage with it, you need to be able to explain the PCA to scholars that are not familiar with using them, in political science at least it is much more common in Europe than in the U.S. The more clearly, simply and concretely you explain it, the more comfortable and confident users will be with using it, and the more they will use it, thus helping the GCI succeed from an impact standpoint."*

This is a great suggestion, and we now plan to include a concrete, step-by-step example that walks readers through how raw country-level data are transformed, combined, and incorporated into the PCA framework. Our objective is to demystify the process, increase transparency, and ensure that users can understand how the index is constructed and why PCA is an appropriate tool for aggregating correlated institutional measures.

Proprietary Data and Formulas

> *"You have likely developed a hugely important resource here, but the mention of "proprietary" and not showing the raw data, will result in far lower usage than if you make it transparent and open. If you make it transparent, all the way down to the compiled raw datasets, so that scholars can really look under the hood, you will likely see citations and usage skyrocket."*

We agree and plan to make every element of the Global Capitalism Index fully transparent and publicly accessible. We are currently building a free public-facing website that will house all data in perpetuity. All underlying datasets, transformations, and aggregation steps will be documented and made available so that researchers can inspect, replicate, and extend the index. Our goal is not to restrict access to the data, but to maximize scholarly engagement, citation, and critical use by allowing users to fully look under the hood.

II. New Business Formation and Growth Subindex

Bankruptcy Law

> *"One thing I heard about while doing fieldwork on my Refugee Entrepreneurship book is the importance of bankruptcy law – some countries, seemed more coming in the Middle East, don't have bankruptcy law, and so if your start up fails, you are personally financially destroyed, and this suppresses business formation in such countries – not sure if you have that under a different sub-index, or a different component, but it didn't seem to be captured here."*

This is very valuable insight, and something we'd like to address as globally comparable datasets grow. Unfortunately, nothing currently exists beyond the World Bank's Doing Business indicators and Enterprise surveys. The World Bank discontinued DB indicators in 2020 due to discovered inconsistencies, and the Enterprise surveys lack the temporal coverage needed for a consistent country-year index. We do currently use the Doing Business Resolving Insolvency indicator in the Strength of Property Rights and Private Ownership subindex and the Market Competition subindex, but intended to avoid the dataset as much as possible. We hoped that the World Bank's Be Ready reports would contain useful information on this front, but the report's constrained geographic scope limits it's applicability for the GCI. That said, the World Bank plans to expand Be Ready's geographic coverage in future years, at which time we will include its datasets in the GCI.

Legal Rights and Entrepreneurship

> *"Also Regarding Government Supporting Policy, in the world I have focused on the last decade, is right to work (refugees), as well as the right to bank, right to start business, the importance of these rights would also extend to women (in some places they are not allowed to work, start businesses, have bank accounts, or own land) So if half a population is limited from doing these things, they will be limited in their ability to form new businesses. You already have a massive amount of data gathered but these examples show how researchers might want to add additional data and key variables to your data to explore new and interesting areas – they'll only be able to do that if they can access your data in unaggregated formats."*

To measure the legal constraints you mention, we incorporate the World Bank's Women, Business and the Law index within the Government Supporting Policy component, which systematically measures gender-based legal barriers to economic participation across countries and over time. While data on refugee-specific rights remain more limited and episodic, the inclusion of this index allows us to account for the broader impact of exclusionary legal regimes on new business formation.

R&D Spending

> *"While you'll likely have the people arguing that some of this could be endogenous, is it that R&D spending is an indicator of a thriving capitalist system? Or do you have a thriving economy, so people spend more on R&D? No way to get around the people that will want to critique on this point I don't think."*

We acknowledge that measures such as R&D spending may be endogenous with economic dynamism. Our approach does not claim strict causal inference at the component level, but rather we include R&D alongside other indicators to proxy whether the environment supports entrepreneurial activity and creative destruction. This aligns with standard practice in comparable global indices[1], where multiple correlated metrics are used to approximate complex institutional and economic phenomena. Principal components analysis accounts for multicollinearity and allows us to combine these datasets while limiting adverse statistical noise.

Concluding Remarks

We are very pleased that you recommend the New Business Formation and Growth subindex for approval as is. While globally comparable data on bankruptcy law remains limited, we now include the World Bank Women, Business and the Law index to capture gender-based legal barriers affecting entrepreneurship. We would be happy to discuss any of these points further, and we hope to continue this productive dialogue about how to strengthen the New Business Formation and Growth subindex and the Global Capitalism Index.

Sincerely,

UVA Democracy and Capitalism Lab

[1] WIPO Global Innovation Index; The Global Competitiveness Report, World Economic Forum.